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Level 4, 2nd Sem / DVOC (Auto. Servicing, PT, AMT)

Subject : Material Science & Materials

Time : 2 Hrs.

M.M. : 50

SECTION-A

Note: Multiple Choice questions. All questions are compulsory. (5x1=5)

- Q.1 Which type of crystal structure does iron have at room temperature
- a) Face-centered cubic (FCC)
 - b) Body-centered cubic (BCC)
 - c) Hexagonal close-packed (HPC)
 - d) Amorphous
- Q.2 Which of the following is an example of synthetic plastic
- a) Wood
 - b) rubber
 - c) bakelite
 - d) cotton
- Q.3 What does annealing do to a metal
- a) increases hardness
 - b) relieves internal stresses and increases ductility
 - c) makes the metal brittle
 - d) increases wear resistance

Q.4 Which of the following materials is commonly used for making tool steel

- a) high carbon steel b) aluminium
c) copper d) lead

Q.5 Which alloying element improves the toughness of steel

- a) silicon b) manganese
c) lead d) gold

SECTION-B

Note: Objective type questions. All questions are compulsory. (5x1=5)

Q.6 Name any one non-ferrous metal.

Q.7 PVC stands for_____.

Q.8 The Bakelite is the trade name of_____.

Q.9 18-4-1 stainless steel contains_____.

Q.10 Write one property of clay wool.

SECTION-C

Note: Short answer type questions. Attempt any six questions out of Eight questions. (6x5=30)

Q.11 Write down the properties and uses of stainless steel.

Q.12 Differentiate between tempering and annealing.

Q.13 Write the uses of high-speed steel.

Q.14 What are the effects of nickel, chromium, and vanadium when added as alloying elements in steel?

Q.15 What are ferrous materials? Write their classification.

Q.16 What is a heat-insulating material? Give some desirable properties of it.

Q.17 Write the advantages of semiconductors (any five).

Q.18 Differentiate between normalizing and tempering.

SECTION-D

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

Q.19 Explain various heat treatment processes.

Q.20 What is glass? Explain its different types.